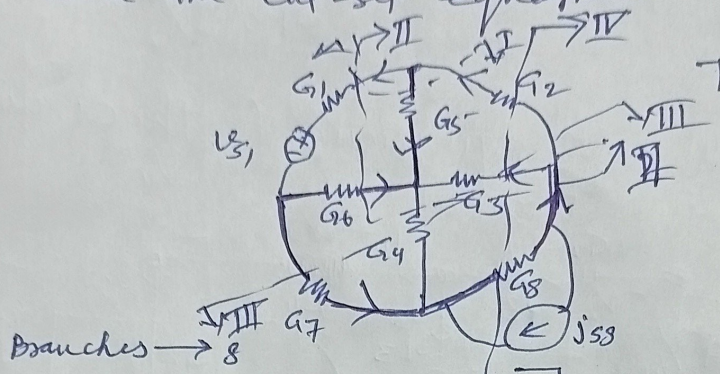
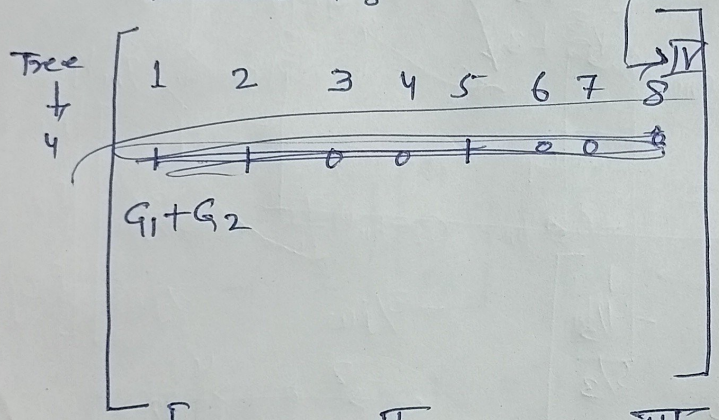
Cut-set analysis

19/08/2025

1. Consider the resistive network shown in Fig-1, Write the cut-set equations.



Tree = {5, 6, 7, 8}

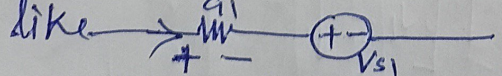


If cut-set reference direction is same the sign is +ve otherwise -ve.

	I	II	III	IV
I	$G_1+G_2+G_3$	$-G_1-G_2$	G_1	G_2
II	$-G_1-G_2$	$G_1+G_2+G_3+G_4+G_5$	$-G_2-G_3-G_4$	$-G_2-G_3$
III	G_2	$-G_2-G_3-G_4$	$G_2+G_3+G_4+G_7$	G_2+G_3
IV	G_2	$-G_2-G_3$	G_2+G_3	$G_2+G_3+G_8$

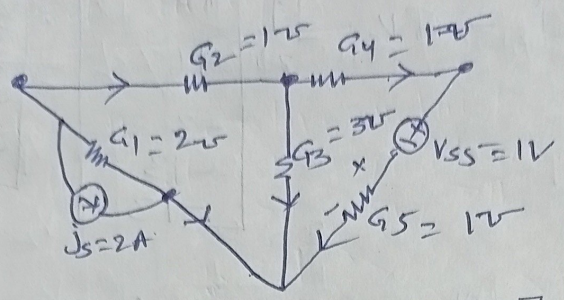
$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix} = \begin{bmatrix} G_1 V_{s1} \\ -G_1 V_{s1} \\ 0 \\ i_{ss} \end{bmatrix}$

- $G_1 V_{s1}$ → If polarities of branch element and source have same like → then $i_{s1} = G_1 V_{s1}$ otherwise vice versa



i_{ss} → If current source reference direction at a point is similar then sign is -ve otherwise +ve.

(2)



- polarity same then sign +ve for voltage source
- Reference direction at a point is opposite then sign +ve for current sources

$$i_s = \mathcal{G} G V_s - \mathcal{G} j_s = \mathcal{G} G \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ +1 \end{bmatrix} - \mathcal{G} \begin{bmatrix} +2 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$J = GV + j_s - GV_s$$

$$= Y_n e = i_s$$

$$Y_n = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 2 \end{bmatrix}; \quad i_s = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

(3)

$$\begin{bmatrix} 3 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 9 & 2 & 1 \\ 2 & 6 & 3 \\ 1 & 3 & 13 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

or,

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} -17 \\ -1 \\ 12 \end{bmatrix}$$

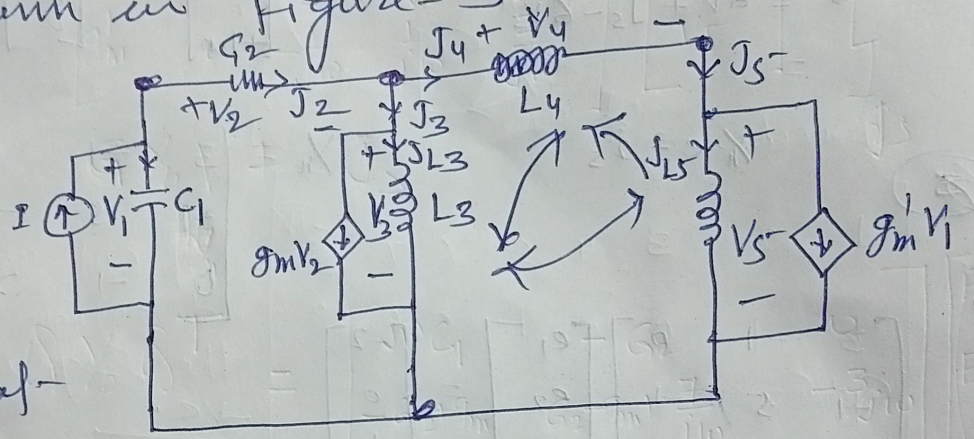
or,

Then $V = \mathcal{G}^T e = \frac{1}{25} \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -17 \\ -1 \\ 12 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} -17 \\ -16 \\ -1 \\ -13 \\ 12 \end{bmatrix}$

$$j = GV + j_s - GV_s = \frac{1}{25}\begin{bmatrix} 16 \\ -16 \\ -3 \\ -13 \\ -13 \end{bmatrix} = \begin{bmatrix} J_1 \\ J_2 \\ J_3 \\ J_4 \\ J_5 \end{bmatrix}$$

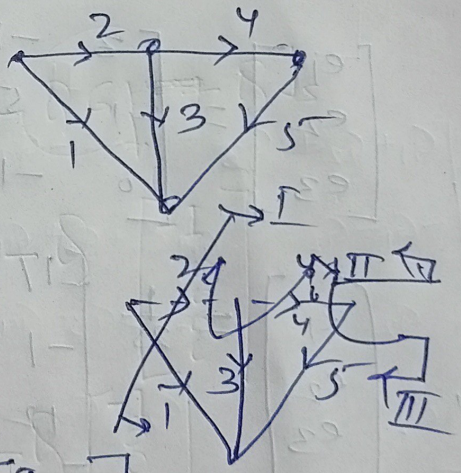
Now, $\begin{bmatrix} J_{L3} \\ J_4 \\ J_{L5} \end{bmatrix}$

Q. Write Cut-set equations of the circuit shown in Figure-3.



Given that-

$$L = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 5/3 & -4/3 \\ 1 & -4/3 & 5/3 \end{bmatrix}$$



Soln:

$$Q = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

$$J_1 = j\omega C_1 V_1 - I \quad \text{(I)}$$

$$J_2 = G_2 V_2 \quad \text{(II)}$$

$$\begin{bmatrix} V_3 \\ V_4 \\ V_5 \end{bmatrix} = j\omega \begin{bmatrix} 1 & -1 & 1 \\ -1 & 5/3 & -4/3 \\ 1 & -4/3 & 5/3 \end{bmatrix} \begin{bmatrix} J_{L3} \\ J_4 \\ J_{L5} \end{bmatrix}$$

Now, $\begin{bmatrix} I_{L3} \\ I_4 \\ I_{L5} \end{bmatrix} = \frac{1}{jw} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 5/3 & -4/3 \\ 1 & -4/3 & 5/3 \end{bmatrix}^{-1} \begin{bmatrix} V_3 \\ V_4 \\ V_5 \end{bmatrix}$

or $\begin{bmatrix} I_{L3} \\ I_4 \\ I_{L5} \end{bmatrix} = \frac{1}{jw} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} V_3 \\ V_4 \\ V_5 \end{bmatrix}$

Since $I_3 = I_{L3} + g_m V_2 \Rightarrow I_{L3} = I_3 - g_m V_2$ — (IV)

and $I_5 = I_{L5} + g_m' V_1 \Rightarrow I_{L5} = I_5 - g_m' V_1$ — (V)

Now, $I_{L3} = \frac{1}{jw}[3V_3 + V_4 - V_5]$ — (VI)

$I_4 = \frac{1}{jw}[V_3 + 2V_4 + V_5]$ — (VII)

$I_{L5} = \frac{1}{jw}[-V_3 + V_4 + 2V_5]$ — (VIII)

~~Substitution steps~~

From eqns (IV) to (VIII), we get

$I_3 = g_m V_2 + \frac{3}{jw} V_3 + \frac{1}{jw} V_4 + \frac{1}{jw} V_5$ — (IX)

$I_4 = \frac{1}{jw} V_3 + \frac{2}{jw} V_4 + \frac{1}{jw} V_5$ — (X)

$I_5 = g_m' V_1 + \frac{1}{jw} V_3 + \frac{1}{jw} V_4 + \frac{2}{jw} V_5$ — (XI)

Now, uniting eqns (I) (II) (IX), (X) & (XI) in matrix form, we get

$$\begin{bmatrix} J_1 \\ J_2 \\ J_3 \\ J_4 \\ J_5 \end{bmatrix} = \begin{bmatrix} j\omega C_1 & 0 & 0 & 0 & 0 \\ 0 & G_2 & 0 & 0 & 0 \\ 0 & g_m & \frac{3}{j\omega} & \frac{1}{j\omega} & -\frac{1}{j\omega} \\ 0 & 0 & \frac{1}{j\omega} & \frac{2}{j\omega} & \frac{1}{j\omega} \\ g_m' & 0 & -\frac{1}{j\omega} & \frac{1}{j\omega} & \frac{2}{j\omega} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} + \begin{bmatrix} -I \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + J_s$$

$$J = G Y_b V + J_s$$

$$Y_n = Q Y_b Q^T = \begin{bmatrix} j\omega C_1 & -G_2 & 0 \\ -G_2 + g_m & G_2 - g_m + \frac{7}{j\omega} & -3/j\omega \\ g_m' & -3/j\omega & 2/j\omega \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix} =$$

$Y_n e = i_s$

E_1
E_2
E_3

$i_s = + Q G V_s + Q j_s$

$= -Q \begin{bmatrix} -I \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$= -Q \begin{bmatrix} -I \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$= Q \begin{bmatrix} I \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} I \\ 0 \\ 0 \end{bmatrix}$ Ans